Linear Algebra Assignment

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1. $\langle p, q \rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + p(t_2)q(t_2)$

$p(t) = 4t - t^2$

At $t_0 = -1,\ t_1 = 0,\ t_2 = +1$

$q(t) = 9 + 3t^2$

$p(t_0) = p(-1) = 4(-1) - (-1)^2$
$= -4 - 1$
$= -5$

$p(t_1) = p(0) = 4(0) - 0^2$
$= 0$

$q(t)_0 = q(-1) = 9 + 3(-1)^2$
$= 9 + 3$
$= 12$

$q(t_1) = q(0)$
$= 9 + 3(0)$
$= 9$

$q(t_2) = q(1) = 9 + 3(1)^2$
$= 12$

$p(t_2) = p(1) = 4(1) - 1^2$
$= 3$

1) $\langle p, q \rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + p(t_2)q(t_2)$

$= (-5)(12) + (0)(9) + (12)(3)$

$= -60 + 36 = -24$

2) $\langle p, p \rangle = [p(t_0)]^2 + [p(t_1)]^2 + [p(t_2)]^2$

$= (-5)^2 + 0^2 + 3^2$

$= 25 + 9$

$= 34$

3) $\langle q, q \rangle = [q(t_0)]^2 + [q(t_1)]^2 + [q(t_2)]^2$

$= (12)^2 + (9)^2 + (12)^2$

$= 144 + 81 + 144$

$= 369$

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2. $\langle f, g \rangle = \int_0^1 f(t)\, g(t)\, dt$

$f(t) = t$

$g(t) = t^2$

$\langle f, g \rangle = \int_0^1 t \cdot t^2\, dt$

$= \int_0^1 t^3\, dt$

$= \left[\frac{t^4}{4}\right]_0^1 \Rightarrow \left[\frac{(1)^4}{4} - \frac{(0)^4}{4}\right] = \frac{1}{4}$

$\therefore \langle f, g \rangle = \frac{1}{4}$

thogonal.

$$v_2 = \left(\frac{1}{\sqrt{10}}\begin{bmatrix}3\\1\end{bmatrix}^T\right)\left(\frac{1}{\sqrt{40}}\begin{bmatrix}-2\\6\end{bmatrix}\right)$$

$$= \left(\frac{1}{\sqrt{400}}\right)\begin{bmatrix}3 & 1\end{bmatrix}\begin{bmatrix}-2\\6\end{bmatrix}$$

$$= \left(\frac{1}{\sqrt{400}}\right)\begin{bmatrix}-6+6\end{bmatrix}$$

$$= 0$$

Inner Product is 0.

$$X = \begin{bmatrix}-6\\3\end{bmatrix}, \quad X = c_1 V_1 + c_2 V_2$$

$$c_1 = \langle x, v_1 \rangle \; ; \; c_2 = \langle x, v_2 \rangle$$

$$c_1 = \left\langle \begin{bmatrix}-6\\3\end{bmatrix}, \frac{1}{\sqrt{10}}\begin{bmatrix}3\\1\end{bmatrix} \right\rangle = \left(\frac{1}{\sqrt{10}}\right)(-18+3) = \frac{-15}{\sqrt{10}} = -\frac{3\sqrt{10}}{2}$$

$$c_2 = \left\langle \begin{bmatrix}-6\\3\end{bmatrix}, \frac{1}{\sqrt{40}}\begin{bmatrix}-2\\6\end{bmatrix} \right\rangle = \frac{1}{\sqrt{40}}(12+18) = \frac{30}{\sqrt{40}} = \frac{3\sqrt{10}}{2}$$

$$x = \frac{-3\sqrt{10}}{2} V_1 + \frac{3\sqrt{10}}{2} V_2$$

2) $$B^T A = \begin{bmatrix} 4 & 3 & -7 \\ -2 & 5 & 6 \\ 7 & 2 & 5 \end{bmatrix} \begin{bmatrix} 3 & 2 & -5 \\ 2 & 1 & 4 \\ 1 & 0 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 12+6-7 & 8+3+0 & -20+12+35 \\ -6+10+6 & -4+5+0 & 10+20-30 \\ 21+4+5 & 14+2+0 & -35+8-25 \end{bmatrix}$$

3) $$= \begin{bmatrix} 11 & 11 & 27 \\ 10 & 1 & 0 \\ 30 & 16 & -52 \end{bmatrix} \Rightarrow Tr(B^T A) = 11+1-52$$

$$= -40$$

$\langle A, B \rangle = -40$

2. 5. $S = \left\{ \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ 1 \end{bmatrix} , \frac{1}{\sqrt{40}} \begin{bmatrix} -2 \\ 6 \end{bmatrix} \right\}$; $x = \begin{bmatrix} -6 \\ 3 \end{bmatrix}$

Orthonormality :- When two vectors u, v are unit vectors and they should be orthogonal inner product.

Basis :- Vectors should span R^2. they are linearly independant

$\|V\| = \sqrt{a^2+b^2}$; $\|V_1\| = \frac{1}{\sqrt{10}} \left(\sqrt{9+1}\right) = \frac{1}{\sqrt{10}} \sqrt{10} = 1$

$\|V_2\| = \frac{1}{\sqrt{40}} \sqrt{4+36} = \frac{1}{\sqrt{40}} \sqrt{40} = 1$

$\|v\| \leq \|(u)\| + \|v\|$

$5.92 \leq 7.03$

Pythogrean Theorem states that Two vectors $\langle u, v \rangle = 0$, are orthogonal

$$\|u+v\|^2 = \|(u)\|^2 + \|v\|^2$$

$$u = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} \quad v = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} \quad u + v = \begin{bmatrix} 4 \\ 4 \\ -2 \end{bmatrix}$$

$$\langle u, v \rangle = 2 \cdot 2 + 5 \cdot (-1) + (-1)(-1)$$

$$= 4 - 5 + 1$$

$$= 0$$

Both are orthogonal

$$\|(u)\| = \sqrt{4 + 25 + 1} \qquad \|v\| = \sqrt{4 + 1 + 1}$$

$$= \sqrt{30} \qquad = \sqrt{6}$$

$$\|(u)\|^2 = 30 \qquad \|v\|^2 = 6$$

$$\|(u+v\| = \sqrt{16 + 16 + 4}$$

$$\|(u+v\|^2 = 36$$

Proved.

5) Swarty :- $\left|\left|\langle f,g\rangle\right|\right| \leq \left|\left|f\right|\right| \left|\left|g\right|\right|$

Fact

1) $\langle f,f\rangle = \int_0^1 t^2 \cdot t^2 \, dt$

$= \int_0^1 t^4$

$= \left[\frac{t^5}{5}\right]_0^1 = \frac{1}{5}$

2) $\langle g,g\rangle = \int_0^1 (1-3t)^2 dt$

$= \int_0^1 (1+9t^2-6t) \, dt$

$= \left[t + \frac{9t^3}{3} - \frac{6t^2}{2}\right]_0^1$

$= \left(1 + \frac{9(1)}{3} - \frac{6(1)}{2}\right) - \left(0 + \frac{9(0)}{3} - \frac{6(0)}{2}\right)$

$= \left(1 + \frac{9}{3} - \frac{6}{2}\right) = 1+1-1$

$= 1$

$\langle f,f\rangle = \frac{1}{5}, \quad \langle g,g\rangle = 1$

$\left|\left|f\right|\right| = \sqrt{\frac{1}{5}}, \quad \left|\left|g\right|\right| = 1$

$\frac{5}{12} \leq \sqrt{\frac{1}{5}} = 0.416 \leq 0.44$

8. $\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$

$\|u+v\|^2 = \langle u+v, u+v \rangle$

$\|u-v\|^2 = \langle u-v, u-v \rangle$

$\langle u+v, u+v \rangle = \|u\|^2 + \|v\|^2 + 2\langle u \cdot v \rangle$

$\langle u-v, u-v \rangle = \|u\|^2 + \|v\|^2 - 2\langle u \cdot v \rangle$

$\langle u+v, u+v \rangle + \langle u-v, u-v \rangle = 2\|u\|^2 + 2\|v\|^2$

9) $u = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 2 \end{bmatrix}, \quad v = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 3 \end{bmatrix}$

Triangle Inequality:- $\|u+v\| \leq \|u\| + \|v\|$

$u + v = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 5 \end{bmatrix}$

$\|u+v\| = \sqrt{9+1+0+25}$
$= \sqrt{35}$

$\|u\| = \sqrt{1+4+1+4}$
$= \sqrt{10}$

$\|v\| = \sqrt{4+1+1+9}$
$= \sqrt{15}$

6) $\left\{ \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}, \begin{bmatrix} 1/3 \\ 2/3 \\ 0 \end{bmatrix} \right\}$ For these to be orthonormal, vectors should be unit, and both be orthogonal (i.e inner product = 0)

$$\|v_1\| = \sqrt{\left(\frac{-2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2}$$

$$= \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} = 1$$

$$\|v_2\| = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + 0}$$

$$= \sqrt{\frac{1}{9} + \frac{4}{9}} = \sqrt{\frac{5}{9}} = \frac{\sqrt{5}}{3}$$

v_2 is not an unit vector, So set is not orthonormal

7) $\langle f, g \rangle = \int_0^1 f(t)\, g(t)$; $f(t) = t^2$; $g(t) = 1 - 3t$

$$\langle f, g \rangle = \int_0^1 t^2 \cdot (1 - 3t)\, dt$$

$$= \int_0^1 t^2 - 3t^3\, dt$$

$$= \int_0^1 t^2\, dt - \int_0^1 3t^3\, dt$$

$$= \left[\frac{t^3}{3}\right]_0^1 - \left[\frac{3t^4}{4}\right]_0^1 \Rightarrow \left(\frac{1}{3} - \frac{0}{3}\right) - \left(\frac{3(1)}{4} - \frac{3(0)}{4}\right)$$

$$= \frac{1}{3} - \frac{3}{4} = \frac{-5}{12}$$

$\langle x, y \rangle = xAy^T$

$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$; $x = \begin{bmatrix} -1 & 3 \end{bmatrix}$; $y = \begin{bmatrix} 1 & -2 \end{bmatrix}$

$y^T = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

$\langle x, y \rangle = xAy^T$

$xAy^T = \begin{bmatrix} -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

$= \begin{bmatrix} -1 & 3 \end{bmatrix} \begin{bmatrix} 1-2 \\ 1-4 \end{bmatrix}$

$= \begin{bmatrix} -1 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ -3 \end{bmatrix}$

$= 1 - 9$

$= -8$

$\langle x, y \rangle = -8$

4) $\langle A, B \rangle = \text{Trace}(B^T A)$

$A = \begin{bmatrix} 3 & 2 & -5 \\ 2 & 1 & 4 \\ 1 & 0 & -5 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -2 & 7 \\ 3 & 5 & 2 \\ -7 & 6 & 5 \end{bmatrix}$; $B^T = \begin{bmatrix} 4 & 3 & -7 \\ -2 & 5 & 6 \\ 7 & 2 & 5 \end{bmatrix}$